# Shangyue Jiang

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## EDUCATION

**University of Connecticut**
Ph.D. in Economics — May 2026 (expected)

**University of Connecticut**
M.S. in Quantitative Economics — Dec 2025 (expected)

**University of Connecticut**
B.A. in Economics — May 2020

## PRIMARY RESEARCH FIELDS

**Applied Microeconomics, Industrial Organization, Production Economics, Education Economics**

## RESEARCH

**The Impact of Weather on U.S. Agricultural Capacity Utilization: A Multiple-Output Nonparametric Analysis** *(Job Market Paper)*
As a significant part of the U.S. economy, the agricultural sector accounted for 5.5% of U.S. gross domestic product and 10.4% of national employment in 2024. Despite its importance, limited research has examined the capacity utilization of the U.S. agricultural industry. This study uses a state-level dataset from 1960 to 2015 and employs a nonparametric data envelopment analysis (DEA) approach to investigate how weather conditions affect agricultural capacity utilization. We estimate capacity utilization using a short-run ray average cost minimization model within a production technology framework that incorporates multiple outputs (Crops, Livestock and products, and Farm related output) produced from multiple variable inputs and multiple fixed inputs (Capital, and Land), following the method introduced by Ray, Walden, and Chen (2021). The findings indicate that most states operated below full capacity and exhibit systematic regional patterns. A second-stage two-way fixed effect model reveals that weather conditions significantly contribute to the shortfall in capacity utilization.

**In-person learning during the pandemic: Student take-up and school-level effects of remote and hybrid instruction on student outcomes** *with Prof. Stephen L. Ross (Proceedings of the National Academy of Sciences (PNAS), published.)*
Student take-up of in-person learning opportunities was low, especially in disadvantaged schools. Accounting for low take-up reveals much larger reductions in pandemic learning loss from in-person enrollment, compared to the effects of district offerings. However, we observe only small gains for individual students from learning in-person, as compared to students at the same school who spent more time learning remotely. Finally, teachers have identified fully synchronous hybrid learning where students are learning fully online when at home as very challenging, and we find that fully synchronous hybrid learning led to larger learning losses. Taken together, these findings suggest that a substantial portion of remote learning losses arose from challenges faced by teachers having to adapt to new learning modes.

**Generalized Leontief Cost Function Analysis of U.S. Ambulatory Care Industry**
The study applied the Generalized Leontief function to examine the production dynamics of the U.S. ambulatory health care service industry between 1987 and 2019, considering a single output and multi-input framework.Utilizing insights from neoclassical duality theory and the Seemingly Unrelated Regression (SUR) estimation method, we determined the elasticity of substitution between different input pairs as well as the price elasticities of input demands. The findings indicate an increasing trend in the degree of complementarity between Labor and IT/Software Capital. While the price elasticities for Labor demand have seen an upward trajectory, those for IT/Software Capital witnessed a decline over the past four decades.

**An Evaluation of the Effectiveness of HomeVisits for Re-Engaging Students Who Were Chronically Absent inthe Era of Covid-19** with Prof. Eric Brunner and Prof. Stephen L. Ross

**A Semi-Parametric Analysis of Non-Discretionary Factors in the Education Sector: Revisiting Connecticut School Districts** (Work in Progress)

## RESEARCH POSITION

**Research Assistant** Dec. 2020 – Current
Prof. Stephen L. Ross and Prof. Eric Brunner, University of Connecticut

- Analyzed how school district responses to the COVID-19 pandemic affected student learning outcomes.
- Investigated the impact of the technical high school system on students' labor market outcomes.
- Designed and implemented empirical analyses and contributed to drafting multiple working papers.

## TEACHING EXPERIENCE

**Instructor of Record, University of Connecticut**

- Econometrics I (Lab Instructor) Fall 2025
- Principles of Macroeconomics (Honor Session) Spring 2025
- Elementary Economic Forecasting Fall 2024

## CONFERENCES AND WORKSHOP PRESENTATIONS (Including scheduled)

- University of Connecticut Economics Department Seminar Fall 2025
- North American Productivity Workshop Summer 2025
- University of Connecticut Economics Department Graduate Student Forum Spring 2025
- University of Connecticut Economics Department Graduate Student Forum Fall 2024

## FUNDING, AWARDS AND FELLOWSHIPS

- Eleanor Bloom Summer Fellowship in Economics, UConn 2023, 2024, 2025
- 2025 Summer Doctoral Dissertation Fellowship, UConn Summer 2025
- Conference Travel Grant, UConn Summer 2025
- 2024 Summer Doctoral Dissertation Fellowship, UConn Summer 2024
- 2023 Summer Doctoral Dissertation Fellowship, UConn Summer 2023
- Timothy A. & Beverly C. Holt Fellowship, UConn 2022
- Graduate School Pre-doctoral Fellowship in Economics, UConn 2021

## REFEREE/REVIEWER

Agribusiness

## OTHER INFORMATION

**Languages:** Mandarin (native), English (fluent)
**Software skills:** Python, STATA, R, Excel VBA

## REFERENCES

- **Subhash C. Ray** (Advisor)- Professor, UConn, subhash.ray@uconn.edu
- **Stephen L. Ross** (Research Assistant Supervisor)- Professor, UConn, stephen.l.ross@uconn.edu
- **Kathleen Segerson** - Professor, UConn, kathleen.segerson@uconn.edu
- **Talia Bar** - Associate Professor, UConn, talia.bar@uconn.edu